

Section 2.2 Experimental Design Instruction for Prompt-Induced Linearity in LLM Negotiation

Prepared from the updated draft and reviewer-oriented robustness requirements

Purpose. This document is a planning blueprint for Section 2.2. It specifies, experiment by experiment, what to run, what hypothesis each experiment tests, what figures or tables to produce, which experiments can be completed with the existing randomized-buyer dataset, and which ones require additional data generation. The design is intentionally conservative and reviewer-facing: it is meant to make the section difficult to attack.

Core empirical claim to support. A substantial portion of the seller behavior induced by prompt-based negotiation personas can be well approximated by a margin-normalized linear policy on a small set of economically interpretable history features. The evidence must therefore show four things at once: good fit, interpretability, invariance across products and time slices, and clear separation across distinct prompt policies.

1. Global protocol to fix before running any experiment

- **Unit of analysis.** The basic observation is one seller response turn t from one negotiation trajectory. All train/validation/test splits must be grouped by full negotiation run, not by individual turns. No two turns from the same negotiation may be split across train and test.
- **Normalization.** For every product with market price M and seller cost v , define normalized seller and buyer prices as $s_t = (S_t - v) / (M - v)$ and $b_t = (B_t - v) / (M - v)$. These normalized variables are the default inputs and outputs for the main experiments.
- **No time features in the main model.** Do not include the round index t , remaining rounds, or any other explicit time variable in the main specification. Time effects are tested later only as a diagnostic robustness check.
- **Cross-validation.** Use grouped 5-fold cross-validation by negotiation run. Keep the same folds across all candidate models and feature sets. Within each fold, average errors at the run level first, then average across runs, so long conversations do not dominate.
- **Main metrics.** Report normalized MAE, normalized RMSE, response-scale R^2 , and dollar MAE = $(M - v) \times$ normalized MAE. For distributional and simulation experiments also report Wasserstein distance or KS distance when appropriate.
- **Uncertainty.** Use bootstrap confidence intervals by resampling complete negotiation runs within each policy. A 1000-bootstrap interval is sufficient.

2. Candidate feature library

Why this section is important. A reviewer can easily object that the phrase 'linear policy' is too flexible unless the feature map is constrained in advance. The feature library below is therefore pre-specified, interpretable, and nested. It allows richer behavior such as asymmetry to low versus high buyer bids, while keeping the fitted model linear in features.

Notation. All features below are computed from normalized history. The default lag order should be tested over τ in $\{1, 2, 3\}$. For any scalar x , define $x^+ = \max(x, 0)$ and $x^- = \max(-x, 0)$.

Family A: basic lag features

- $A1(\tau=1)$: $[1, b_t, s_{t-1}]$
- $A2(\tau=2)$: $[1, b_t, s_{t-1}, b_{t-1}, s_{t-2}]$

- $A3(\tau=3)$: $[1, b_t, s_{t-1}, b_{t-1}, s_{t-2}, b_{t-2}, s_{t-3}]$

Family B: sign-split level features

- Replace current and lagged buyer bids by positive and negative parts: $b_t^+, b_t^-, b_{t-1}^+, b_{t-1}^-$, ...
- Interpretation: a buyer bid above cost and a buyer bid below cost need not influence the seller symmetrically.
- Recommended default at $\tau=2$: $[1, b_t^+, b_t^-, s_{t-1}, b_{t-1}^+, b_{t-1}^-, s_{t-2}]$

Family C: movement and concession features

- Buyer movement: $\Delta b_t = b_t - b_{t-1}$; use $(\Delta b_t)^+$ and $(\Delta b_t)^-$
- Seller movement: $\Delta s_{t-1} = s_{t-1} - s_{t-2}$; use $(\Delta s_{t-1})^+$ and $(\Delta s_{t-1})^-$ if desired
- Interpretation: some policies respond more to upward buyer movement than to downward or stagnant movement.

Family D: gap and reciprocal-pressure features

- Current gap: $g_t = s_{t-1} - b_t$, with g_t^+, g_t^- if needed
- Previous gap: $g_{t-1} = s_{t-2} - b_{t-1}$
- Buyer seriousness ratio: $r_t = (b_t - b_{t-1}) / (s_{t-1} - b_{t-1} + \text{small epsilon})$, clipped to a reasonable range
- Interpretation: the seller may react to whether the buyer is closing the gap, not only to the price level itself.

Feature selection rule. The paper should not report dozens of arbitrary feature combinations. Instead, compare Family A, Family B, Family B+C, and Family B+C+D, each at $\tau = 1, 2, 3$. Choose one common final feature set for the main text by the one-standard-error rule on cross-validated MAE. The expectation is that sign-split features will matter, but the final chosen family should remain small and interpretable.

3. Candidate prediction models

- Mean predictor
- Persistence predictor: predict $s_t = s_{t-1}$
- OLS or ridge regression on normalized response s_t
- Main linear model: ridge regression on normalized response s_t using the selected feature family
- If you later return to the sigmoid / logit formulation, keep the same feature families and compare it explicitly against response-space linear regression
- GAM with splines on the same inputs
- Gradient-boosted trees on the same inputs
- Small MLP on the same inputs

Regularization. Use ridge as the default linear estimator. Standardize all non-binary features inside each training fold. Select the penalty by inner cross-validation.

4. Experiment-by-experiment plan

Experiment 1. Dataset audit and boundedness check

Data requirement: Existing data only.

Goal: Verify that the randomized dataset is clean and that the linear-policy setup is empirically plausible before fitting any model.

What to run.

- For each policy and product, count runs, total seller turns, average turns per run, min and max normalized seller response, fraction of seller prices below cost, fraction above market, and fraction of exact acceptances where $S_t = B_t$.
- If there are any parsing failures or malformed price outputs, record them and exclude them transparently.

What this tests.

- Does the data mostly respect the economically feasible range?
- How much of the dataset is standard counteroffer behavior versus immediate acceptance behavior?

What to show.

- Table E1: dataset summary by policy and by product.
- Histogram or violin plot of normalized seller prices by policy.

Planning note. This table should appear early in the section or appendix. It prevents later objections about data quality.

Experiment 2. Feature-family and model comparison

Data requirement: Existing data only.

Goal: Test whether a small linear model is actually good, and whether richer feature engineering such as sign-split buyer variables materially helps.

What to run.

- For each policy, fit all candidate models on all candidate feature families, using grouped 5-fold CV.
- At minimum compare: Family A, Family B, Family B+C, Family B+C+D.
- For each family, compare persistence, linear ridge, GAM, boosted trees, and MLP.
- Use the same folds across all models and report out-of-fold predictions.

What this tests.

- How far does a linear model outperform trivial baselines?
- Do sign-split buyer features b_t^+ and b_t^- improve fit relative to raw b_t ?
- How close is the linear model to nonlinear competitors?

What to show.

- Table E2: cross-validated MAE, RMSE, dollar MAE, and R^2 for every policy $\times$ model $\times$ feature family.
- Figure E1: dot plot of MAE by model family averaged over policies.
- Figure E2: improvement from Family A to Family B and beyond, shown as paired differences.

Planning note. This is the core experiment. The final feature family for the main paper should be selected here.

Experiment 3. Reader-visible fit diagnostics

Data requirement: Existing data only.

Goal: Make the predictive quality visually obvious, not just numerically reported.

What to run.

- Using the selected linear model from Experiment 2, collect out-of-fold predictions for all seller turns.
- Prepare prediction diagnostics separately by policy and also pooled across policies.

What this tests.

- Do predicted seller proposals track actual seller proposals closely enough to be convincing?
- Are errors concentrated in a few edge cases, or broadly small across the state space?

What to show.

- Figure E3: predicted versus actual normalized seller prices with a 45-degree reference line.
- Figure E4: residual histogram and residual versus fitted plot.
- Figure E5: binned calibration plot or conditional mean plot of predicted versus actual seller response.

Planning note. A reviewer should be able to see the model quality immediately from these plots.

Experiment 4. Variable importance and explicit strategy interpretation

Data requirement: Existing data only.

Goal: Explain which variables matter, and turn coefficients into an interpretable description of each prompted strategy.

What to run.

- Fit the selected linear model separately for each policy using the full randomized dataset.
- Standardize all continuous features before fitting, then report standardized coefficients.
- Compute three complementary importance measures: standardized coefficient magnitude, drop-column delta-MAE, and permutation importance on held-out folds.
- For sign-split features, combine and compare the influence of b_{t^+} versus b_{t^-} and of positive versus negative buyer movement.
- Write a short interpretation paragraph for each policy based on the sign and magnitude patterns.

What this tests.

- Which variables dominate the seller response?
- Does the policy react more to current buyer level, previous seller anchor, buyer movement, or remaining gap?
- Are low bids punished asymmetrically relative to serious buyer movement?

What to show.

- Figure E6: coefficient heatmap by policy and feature.
- Figure E7: drop-column importance plot by policy.
- Table E3: textual interpretation of each strategy, for example 'PVD is anchor-heavy and weakly reactive to low buyer bids' or 'RFK responds strongly to positive buyer movement and current gap closure'.

Planning note. This experiment is essential because the paper claims to recover a policy, not merely a black-box predictor.

Experiment 5. Lag sufficiency

Data requirement: Existing data only.

Goal: Show that the seller policy depends on a short recent history rather than a long memory.

What to run.

- For the chosen feature family, fit the linear model with $\tau = 1, 2, 3$.
- Report predictive performance by τ for every policy.

What this tests.

- Does performance saturate quickly with short lag order?
- Can one justify a common lag order across policies?

What to show.

- Figure E8: MAE versus lag order for each policy.
- Table E4: selected τ by the one-standard-error rule.

Planning note. If $\tau = 2$ is nearly optimal everywhere, the final presentation becomes much cleaner.

Experiment 6. Margin-normalization ablation

Data requirement: Existing data only.

Goal: Show that margin normalization is not just cosmetic; it is what enables transfer across products.

What to run.

- Using the selected linear feature family and linear estimator, fit three variants: raw-dollar features and response, market-only scaling, and full margin normalization.
- Evaluate with leave-one-product-out transfer: train on four products and test on the held-out fifth.

What this tests.

- Does full normalization materially improve held-out-product performance?
- Does normalization reduce cross-product instability?

What to show.

- Table E5: held-out-product MAE and RMSE under each scaling choice.
- Figure E9: paired plot of leave-one-product-out error for raw, market-only, and full margin normalization.

Planning note. This experiment is one of the strongest defenses of the conceptual setup in Section 2.1.

Experiment 7. Cross-product stability of recovered models

Data requirement: Existing data only.

Goal: Test whether the same prompted strategy yields similar recovered policies across distinct vehicles.

What to run.

- For each policy, fit one model per product and one pooled model across all products.

- Compare them both by coefficients and by functional predictions on a shared reference grid of normalized states.
- Also compute train-on-product-i, test-on-product-j transfer performance.

What this tests.

- Are product-specific recovered models similar after normalization?
- Is the pooled policy a good summary of product-level policies?

What to show.

- Figure E10: coefficient heatmap across products for each policy.
- Figure E11: cross-product transfer matrix.
- Table E6: pairwise coefficient correlations and functional distances.

Planning note. Functional distance is important because coefficient instability alone can be misleading under collinearity.

Experiment 8. Time-homogeneity diagnostic

Data requirement: Existing data only.

Goal: Check the assumption that the seller policy is approximately time-homogeneous when the true deadline is hidden.

What to run.

- Split the data into early, middle, and late seller turns. A reasonable partition is turns 2–5, 6–10, and 11–16, adjusted to respect the chosen lag.
- Fit separate models in each bucket and compare them.
- As an ablation, add a time feature or bucket dummy to the linear model and record the incremental gain.

What this tests.

- Are early, middle, and late recovered policies similar?
- Does explicitly adding time provide only a small improvement?

What to show.

- Figure E12: heatmap of coefficient differences or functional distances across time buckets.
- Table E7: MAE for baseline model versus baseline plus time indicator.

Planning note. This is how you defend the exclusion of round-based features.

Experiment 9. Cross-policy distinctiveness

Data requirement: Existing data only.

Goal: Show that the same structural model class fits multiple prompts, but the learned parameters are genuinely policy-specific.

What to run.

- Fit one selected linear model per policy.
- Evaluate each model on every other policy's data using the same normalized state variables.

What this tests.

- Does a matched-policy model predict its own policy much better than other policies?
- Are some prompts closer together than others in functional space?

What to show.

- Figure E13: policy-to-policy transfer heatmap.
- Figure E14: clustering or low-dimensional embedding of recovered policies.
- Table E8: matched-policy error versus average mismatched-policy error.

Planning note. This experiment is crucial because it shows you are not simply fitting one generic dealer rule.

Experiment 10. Buyer-template sensitivity check

Data requirement: Existing data only, assuming buyer template identity can be recovered from the message string or logs.

Goal: Rule out the objection that the recovered seller policy is really reacting to wording templates rather than prices.

What to run.

- Fit the selected model with no template controls.
- Fit a nuisance-augmented version that includes buyer-template dummy variables.
- Run leave-one-template-out testing.

What this tests.

- How much predictive gain comes from template identity?
- Does the price-history model transfer across all buyer templates?

What to show.

- Table E9: delta-MAE after adding template dummies.
- Figure E15: leave-one-template-out error across templates.

Planning note. If template effects are tiny, this is a very useful rebuttal to semantics-based criticisms.

Experiment 11. Rollout fidelity of the recovered policy

Data requirement: Existing data only.

Goal: Show that the recovered policy is useful as a full negotiation surrogate, not only as a one-step regressor.

What to run.

- Fit the selected model on the randomized dataset.
- Simulate seller responses against the same random-buyer mechanism used in data generation.
- Compare simulated and real negotiations on trajectory summaries.

What this tests.

- Does the recovered policy reproduce seller ask trajectories, concession patterns, agreement turn distribution, and final price distribution?

What to show.

- Figure E16: real versus simulated mean normalized seller ask by turn.
- Figure E17: final normalized price distribution, real versus simulated.
- Table E10: summary-statistic distances between real and simulated rollouts.

Planning note. This makes the section stronger because it connects one-step fit to strategic behavior.

Experiment 12. Transfer to LLM-buyer negotiations

Data requirement: Requires additional data generation unless the practical-scenario dataset already exists.

Goal: Test whether seller strategies estimated from randomized buyer bids remain useful when the buyer is also an LLM agent.

What to run.

- Train seller models only on the randomized-buyer dataset.
- Collect a test set where the buyer is an LLM using the practical scenario protocol.
- Recommended minimum: 50 runs per product per policy. Stronger: 100 runs per product per policy.
- Optionally include both mirrored buyer personas and mismatched buyer personas.

What this tests.

- Does the recovered seller model still predict well under a new state distribution induced by an agentic buyer?
- How much does performance degrade relative to the randomized-buyer test set?

What to show.

- Table E11: randomized-buyer versus LLM-buyer test error.
- Figure E18: distribution shift plot for key state variables and prediction error by buyer type.

Planning note. Mark this clearly as additional data generation.

Experiment 13. Backend robustness

Data requirement: Requires additional data generation.

Goal: Show that the structural conclusion is not unique to DeepSeek Reasoner.

What to run.

- Repeat the randomized-buyer data generation pipeline with a second backend, ideally GPT-4o or another strong model.
- Keep prompts, products, templates, and sampling protocol unchanged.

What this tests.

- Does the same linear feature family remain competitive on another backend?
- Are recovered policies qualitatively similar across backends?

What to show.

- Table E12: core model-comparison table by backend.
- Figure E19: coefficient or functional similarity between backends for each prompt policy.

Planning note. This is one of the most important new-data experiments for reviewer confidence.

Experiment 14. Controlled state-sweep visualization

Data requirement: Requires additional data generation.

Goal: Provide the single most convincing visual demonstration of the recovered policy.

What to run.

- Choose several anchor negotiation histories from real data.
- Freeze the full history and vary only one current-state variable, usually the buyer's current normalized bid.
- For each anchor state, query the seller at 11 evenly spaced buyer bids within a feasible range and repeat 5 times per bid.

What this tests.

- Does the seller response curve match the recovered linear policy visually?
- Do different prompts produce visibly different response curves?

What to show.

- Figure E20: state-sweep panels showing empirical mean seller response and fitted linear response curve.
- Optional companion plot: same sweep in transformed feature coordinates if you use sign-split or gap features.

Planning note. This is a very high-value figure for the main text.

Experiment 15. Negative controls from threshold-like prompts

Data requirement: Requires additional data generation.

Goal: Show that the paper is not cherry-picking only linear-friendly prompt styles.

What to run.

- Instantiate at least two additional seller prompts that are expected to be harder for a smooth linear policy, such as grim-trigger and corridor policies from the prompt taxonomy.
- Generate a smaller randomized dataset for these prompts, for example 2 products $\times$ 50 runs each.

What this tests.

- Do nonlinear models outperform the selected linear model more substantially on these threshold-like prompt policies?

What to show.

- Table E13: linear versus nonlinear performance on negative-control prompts.
- Figure E21: state-sweep plots revealing visible kinks or punishment thresholds.

Planning note. This is one of the best ways to make the positive results believable.

Experiment 16. Noise-floor estimation by repeated identical states

Data requirement: Requires additional data generation.

Goal: Separate irreducible LLM stochasticity from model misspecification.

What to run.

- Sample around 100 negotiation states from held-out data, balanced across policies and products.
- Replay each exact same history to the seller 10 times.
- Measure within-state variance and compare it with model error.

What this tests.

- How much of the residual error is due to output randomness rather than a wrong structural model?

What to show.

- Table E14: average within-state standard deviation by policy.
- Figure E22: model absolute error versus repeated-query standard deviation.

Planning note. This is especially useful if reviewers push on residual variance.

5. Recommended main-text versus appendix allocation

- Main text: Experiments 2, 3, 4, 6, 7, 8, 9, and 14.
- Appendix: Experiments 1, 5, 10, 11, 12, 13, 15, and 16.
- If space is tight, keep at least one visual fit diagnostic, one variable-importance figure, one normalization transfer figure, one cross-policy transfer figure, and one controlled state-sweep figure in the main text.

6. Minimum package versus strongest package

Minimum package using existing data only. Experiments 1 through 11 are enough to make the section substantially stronger, with special emphasis on Experiments 2, 4, 6, 8, and 9.

Strongest package that is hard to attack. Add Experiments 12 through 16, in the following priority order: controlled state-sweep, backend robustness, negative controls, LLM-buyer transfer, and noise-floor estimation.

7. Final writing guidance for Section 2.2

- Describe the selected feature family explicitly in the main text. Do not leave the feature map as an unspecified black box.
- If sign-split features matter, explain them economically: low buyer bids below cost, serious positive buyer movement, and gap closure need not affect the seller symmetrically.
- Use the variable-importance experiment to write one concise strategy interpretation paragraph per policy.
- Do not rely only on prediction tables. Include at least one figure that a reviewer can understand at a glance.
- If nonlinear models win only marginally, say so directly. If they win substantially for some prompts, report that honestly and use it to delimit the claim.
- Whenever a robustness experiment uses new data, label it explicitly as a new-generation experiment.